

Graphs, graphons, and their eigenvalues

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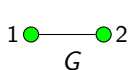
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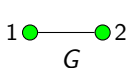
- Consider a graph $G = (V, E)$ with $|V| = n \geq 2$.
- $A :=$ the adjacency matrix of G .
- The eigenvalues of the matrix A : $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$.



$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

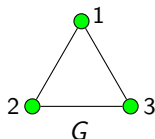
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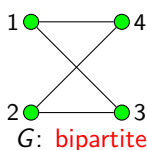
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$$\lambda_1 = 2, \lambda_2 = \lambda_3 = -1$$

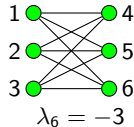
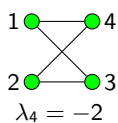


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$$\lambda_1 = 2, \lambda_2 = \lambda_3 = 0, \lambda_4 = -2$$

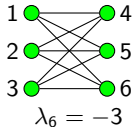
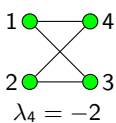
- Assume that every vertex of G has degree d , and G is connected.

G is bipartite $\iff \lambda_n = -d$.



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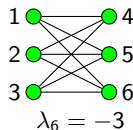
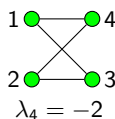


$\frac{\lambda_n - (-d)}{d}$ is “large” $\iff G$ is “away from being bipartite”.

Measuring bipartiteness

- Assume that every vertex of G has degree d , and G is connected.

$$G \text{ is bipartite} \iff \lambda_n = -d.$$



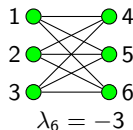
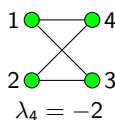
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- Bipartiteness ratio β_G : a “combinatorial” measure of “lack” of bipartiteness.

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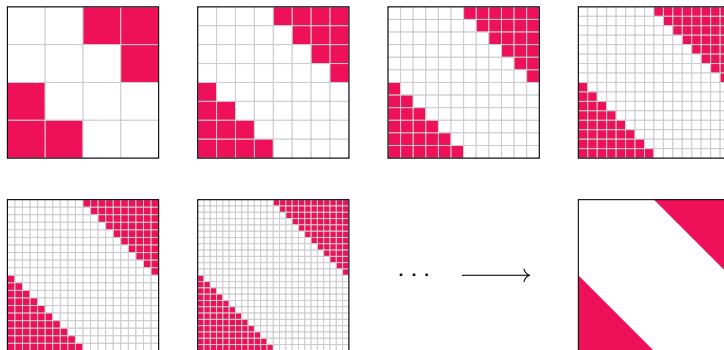
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The dual Cheeger–Buser inequality for graphs (Trevisan, Bauer–Jost)

$$\frac{\beta_G^2}{2} \leq \frac{d + \lambda_n}{d} \leq 2\beta_G$$

Limits of sequences of graphs



The “limits” of sequences of graphs can be thought of as certain “nice” functions from $[0, 1]^2$ to $[0, 1]$, called as **graphons**.

Theorem (Pokharanakar)

- W : a **connected** graphon,
- β_W : the **bipartiteness ratio** of W
- λ_W : an **analog of λ_n** for W

The dual Cheeger–Buser inequality for graphons

$$\frac{\beta_W^2}{2} \leq 2 - \lambda_W \leq 2\beta_W$$

Thank you